

INDU 211: Introduction to Production & Manufacturing Systems

Problem Solution Guide 05: Traveling Salesperson Problem (Warehouse Forklift Routing)

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[!NOTE]

- **YouTube Video Tutorial:** [Watch Video 5: INDU 211 - TSP Problem](#)
- **Video ID:** [ayqA56IHMZ0](#) · **Duration:** 15:24
- **Target Exam Scope:** 🎯 **MIDTERM EXAM (Week 6, Chapters 1–5 & 7)**
- **Exact Exam Match:** **Concordia 2020 Midterm Exam Problem 2 (18 Marks)**
- **Curriculum Context:** Chapter 5 — Materials Handling, Distribution & Routing

1. Problem Description & Warehouse Asymmetric Distance Matrix

A warehouse forklift truck is stationed at parking depot **P** and must visit five production departments (**A, B, C, D, and E**) once per day to deliver batch parts, returning to **P** at the conclusion of the shift. All warehouse corridors are strictly **one-way**, producing an asymmetric distance matrix:

Warehouse Asymmetric Distance Matrix (d_{ij} in meters):

From \ To	P	A	B	C	D	E
P	—	15	18	20	25	13
A	24	—	12	16	13	14
B	19	14	—	18	15	17
C	22	16	21	—	19	15
D	20	17	19	20	—	12
E	18	9	15	13	16	—

2. Step-by-Step Problem Solutions

Part 2.1: Nearest Neighbor (NN) Heuristic Solution

- **Leg 1 (Start at Depot P):** Distances from P to A, B, C, D, E : 15, 18, 20, 25, **13**.
 - Minimum distance is to **E** ($d_{PE} = 13$).
 - Tour so far: $P \rightarrow E$. Visited: P, E . Remaining: A, B, C, D .
- **Leg 2 (Move from E):** Distances from E to A, B, C, D : **9**, 15, 13, 16.
 - Minimum distance is to **A** ($d_{EA} = 9$).
 - Tour so far: $P \rightarrow E \rightarrow A$. Visited: P, E, A . Remaining: B, C, D .
- **Leg 3 (Move from A):** Distances from A to B, C, D : **12**, 16, 13.
 - Minimum distance is to **B** ($d_{AB} = 12$).
 - Tour so far: $P \rightarrow E \rightarrow A \rightarrow B$. Visited: P, E, A, B . Remaining: C, D .
- **Leg 4 (Move from B):** Distances from B to C, D : 18, **15**.
 - Minimum distance is to **D** ($d_{BD} = 15$).
 - Tour so far: $P \rightarrow E \rightarrow A \rightarrow B \rightarrow D$. Visited: P, E, A, B, D . Remaining: C .
- **Leg 5 (Move from D to Last Node C):** Only remaining unvisited node is **C**:
$$d_{DC} = 20$$
 - Tour so far: $P \rightarrow E \rightarrow A \rightarrow B \rightarrow D \rightarrow C$.
- **Leg 6 (Return from C to Depot P):** Return to depot P:

$$d_{CP} = 22$$

Total Distance for Nearest Neighbor Tour:

$$\text{Tour 1 : } P \rightarrow E \rightarrow A \rightarrow B \rightarrow D \rightarrow C \rightarrow P$$

$$\text{Total Distance} = 13 + 9 + 12 + 15 + 20 + 22 = \mathbf{91} \text{ meters}$$

Part 2.2: Second-Nearest Neighbor Variant Solution

- **Rule:** Choose the **second nearest neighbor** on the first step out of depot P, then revert strictly to the nearest neighbor rule for all remaining stops.
- **Leg 1 (Start at Depot P):** Distances from P: E (13), A (15), B (18), C (20), D (25).
 - 1st nearest: E (13).
 - **2nd nearest: A (15).**
 - Tour so far: $P \rightarrow A$. Visited: P, A . Remaining: B, C, D, E .

- **Leg 2 (Move from A):** Unvisited B, C, D, E : **12**, 16, 13, 14 $\implies$ Pick **B** ($d_{AB} = 12$).
 - Tour so far: $P \rightarrow A \rightarrow B$.
- **Leg 3 (Move from B):** Unvisited C, D, E : 18, **15**, 17 $\implies$ Pick **D** ($d_{BD} = 15$).
 - Tour so far: $P \rightarrow A \rightarrow B \rightarrow D$.
- **Leg 4 (Move from D):** Unvisited C, E : 20, **12** $\implies$ Pick **E** ($d_{DE} = 12$).
 - Tour so far: $P \rightarrow A \rightarrow B \rightarrow D \rightarrow E$.
- **Leg 5 (Move from E to Last Node C):** Only unvisited node is **C**: $d_{EC} = 13$.
 - Tour so far: $P \rightarrow A \rightarrow B \rightarrow D \rightarrow E \rightarrow C$.
- **Leg 6 (Return from C to Depot P):** Return to depot P: $d_{CP} = 22$.

Total Distance for Variant Tour:

$$\text{Tour 2 : } P \rightarrow A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow P$$

$$\text{Total Distance} = 15 + 12 + 15 + 12 + 13 + 22 = \mathbf{89 \text{ meters}}$$

Part 2.3: Mathematical Evaluation of Heuristic Optimality

Question:

Are the solutions found in 2.1 (91 m) or 2.2 (89 m) optimal? Explain why or why not.

- **Evaluation: NEITHER solution is guaranteed to be optimal.**
- **Engineering Justification:**
 1. **Greedy Myopic Behavior:** The nearest neighbor method is a greedy heuristic that selects locally optimal short legs early, but completely ignores global network structure.
 2. **The End-of-Tour Penalty:** By making short hops early, the forklift is forced into extremely long, suboptimal penalty legs at the end to close the loop ($D \rightarrow C = 20 \text{ m}$ and $C \rightarrow P = 22 \text{ m}$).
 3. **Direct Empirical Proof:** The fact that **Tour 2 (89 m)** is shorter than **Tour 1 (91 m)** proves that choosing the local minimum at step 1 (E at 13 m) produced a

worse

 global tour (91 m) than choosing the longer initial leg (A at 15 m).

Part 2.4: Search Space & Combinatorial Explosion

Question:

A student builds a "Total Search" app that enumerates all possible routes. Evaluate its optimality and compute the search space for 6 and 21 locations.

- **2.4.1 Is Total Search Guaranteed Optimal?**

- **Yes.** Exhaustive total search evaluates every single feasible closed tour in the finite solution space and selects the minimum. It is mathematically guaranteed to find the true global optimum.

- **2.4.2 Total Number of Routes for 6 Locations (Depot P + 5 Departments):**

$$\text{Total Permutations} = (n - 1)! = (6 - 1)! = 5! = \mathbf{120} \text{ possible tours}$$

- **2.4.3 Total Number of Routes for 20 Additional Departments (Depot P + 20 Departments = 21 Locations):**

$$\text{Total Permutations} = (21 - 1)! = 20! \approx \mathbf{2.433 \times 10^{18}} \text{ tours}$$

$$(Or \text{ if } 20 \text{ locations total including } P: (20 - 1)! = 19! \approx \mathbf{1.216 \times 10^{17}} \text{ tours})$$

- At 1 billion route evaluations per second, exhaustive total search would take **77.1 years**, proving the **Combinatorial Explosion** of NP-hard problems and why heuristics are essential in practice.